

## CAMBRAI EPINOY, France

WMO: 070170

Lat: 50.222N

Lon: 3.154E

Elev: 78

StdP: 100.39

Time Zone: 1.00 (EUC)

Period: 99-12

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -6.2       | -4.1 | -8.9                          | 1.8 | -4.7 | -6.5 | 2.2 | -3.3 | 15.0                  | 9.5  | 13.5 | 8.9  | 3.4                   | 50   | 0.606 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 9.6                    | 29.4            | 20.1 | 27.2 | 19.2 | 25.3 | 18.4 | 21.1                | 27.5 | 20.0 | 25.7 | 19.1 | 24.0 | 4.2                  | 170  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 18.7                            | 13.7 | 23.4 | 18.0 | 13.0 | 22.2 | 17.2 | 12.4 | 21.4 | 61.2          | 27.9 | 57.5 | 25.7 | 54.6 | 24.2 | 24.9              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |      |     |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|------|-----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |      |     |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |      |     |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |      |     |
| (4) 11.7          | 10.3 | 9.2 | DB                         | -8.6 | 33.1               | 2.7 | 2.2                                                | -10.6 | 34.6       | -12.1 | 35.9       | -13.6 | 37.2       | -15.6 | 38.8 | (4) |
| (5)               |      |     | WB                         | -8.9 | 22.9               | 2.7 | 1.0                                                | -10.9 | 23.7       | -12.4 | 24.3       | -13.9 | 24.8       | -15.9 | 25.6 | (5) |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |  |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|--|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 10.8        | 3.4   | 4.3   | 7.1   | 10.2  | 13.5  | 16.3  | 18.1  | 18.1  | 15.8  | 11.7  | 7.3   | 3.6   | (6)  |  |
|                                                                        | DBStd       | 6.32        | 4.18  | 3.83  | 3.48  | 3.41  | 3.14  | 3.00  | 2.92  | 2.65  | 2.86  | 3.33  | 3.34  | 3.92  | (7)  |  |
|                                                                        | HDD10.0     | 824         | 205   | 161   | 101   | 38    | 6     | 0     | 0     | 0     | 0     | 21    | 92    | 200   | (8)  |  |
|                                                                        | HDD18.3     | 2834        | 462   | 393   | 348   | 245   | 153   | 77    | 42    | 36    | 86    | 204   | 331   | 457   | (9)  |  |
|                                                                        | CDD10.0     | 1123        | 2     | 2     | 11    | 43    | 114   | 188   | 250   | 251   | 175   | 75    | 11    | 2     | (10) |  |
|                                                                        | CDD18.3     | 90          | 0     | 0     | 0     | 0     | 3     | 14    | 33    | 28    | 11    | 1     | 0     | 0     | (11) |  |
|                                                                        | CDH23.3     | 870         | 0     | 0     | 0     | 9     | 47    | 158   | 306   | 254   | 91    | 5     | 0     | 0     | (12) |  |
|                                                                        | CDH26.7     | 222         | 0     | 0     | 0     | 0     | 4     | 38    | 85    | 80    | 14    | 1     | 0     | 0     | (13) |  |
| (14) Wind                                                              | WSAvg       | 4.6         | 5.5   | 5.2   | 4.9   | 4.5   | 4.3   | 3.9   | 4.1   | 4.0   | 4.2   | 4.8   | 5.1   | 5.3   | (14) |  |
| (15) Precipitation                                                     | PrecAvg     | 729         | 59    | 53    | 50    | 47    | 59    | 65    | 67    | 71    | 55    | 65    | 66    | 73    | (15) |  |
|                                                                        | PrecMax     | 995         | 145   | 107   | 169   | 125   | 124   | 127   | 112   | 167   | 124   | 142   | 119   | 182   | (16) |  |
|                                                                        | PrecMin     | 558         | 6     | 12    | 5     | 2     | 6     | 28    | 24    | 11    | 12    | 13    | 19    | 22    | (17) |  |
|                                                                        | PrecStd     | 111         | 29    | 27    | 32    | 28    | 31    | 29    | 26    | 38    | 30    | 29    | 28    | 38    | (18) |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 12.9  | 13.7  | 19.0  | 24.6  | 26.8  | 30.7  | 31.7  | 33.1  | 28.6  | 22.2  | 16.1  | 13.0  | (19) |  |
|                                                                        |             | MCWB        | 11.4  | 10.5  | 12.0  | 15.6  | 18.8  | 21.1  | 20.5  | 21.6  | 19.2  | 17.5  | 14.0  | 11.8  | (20) |  |
|                                                                        | 2%          | DB          | 11.7  | 11.9  | 16.4  | 21.8  | 24.6  | 27.3  | 29.3  | 29.3  | 26.0  | 19.4  | 14.4  | 11.8  | (21) |  |
|                                                                        |             | MCWB        | 10.6  | 9.9   | 11.7  | 14.4  | 17.0  | 19.1  | 20.1  | 20.2  | 18.7  | 16.2  | 12.9  | 10.8  | (22) |  |
|                                                                        | 5%          | DB          | 10.5  | 10.8  | 14.3  | 18.6  | 22.5  | 24.9  | 26.9  | 26.4  | 23.5  | 17.7  | 13.2  | 10.6  | (23) |  |
|                                                                        |             | MCWB        | 9.4   | 9.3   | 10.8  | 12.8  | 16.2  | 18.2  | 19.1  | 19.0  | 17.8  | 15.1  | 11.9  | 9.7   | (24) |  |
|                                                                        | 10%         | DB          | 9.3   | 9.7   | 12.6  | 16.3  | 20.0  | 22.6  | 24.6  | 23.7  | 21.1  | 16.4  | 12.1  | 9.4   | (25) |  |
|                                                                        |             | MCWB        | 8.3   | 8.4   | 10.0  | 11.8  | 15.0  | 17.0  | 18.3  | 17.7  | 16.6  | 14.3  | 10.9  | 8.5   | (26) |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 11.9  | 11.4  | 13.4  | 16.7  | 19.9  | 22.5  | 21.9  | 22.1  | 20.9  | 18.1  | 14.2  | 12.1  | (27) |  |
|                                                                        |             | MCDB        | 12.6  | 12.7  | 16.3  | 23.3  | 25.4  | 28.8  | 28.3  | 30.1  | 25.9  | 21.1  | 15.4  | 12.7  | (28) |  |
|                                                                        | 2%          | WB          | 10.7  | 10.4  | 12.4  | 14.8  | 17.9  | 20.1  | 21.0  | 20.9  | 19.5  | 16.6  | 13.1  | 11.0  | (29) |  |
|                                                                        |             | MCDB        | 11.5  | 11.5  | 15.2  | 19.5  | 22.4  | 25.7  | 27.5  | 27.7  | 24.5  | 18.8  | 14.1  | 11.7  | (30) |  |
|                                                                        | 5%          | WB          | 9.5   | 9.5   | 11.4  | 13.5  | 16.6  | 18.9  | 19.8  | 19.7  | 18.3  | 15.6  | 12.1  | 9.7   | (31) |  |
|                                                                        |             | MCDB        | 10.4  | 10.5  | 14.0  | 17.7  | 21.3  | 23.8  | 25.3  | 25.1  | 22.3  | 17.4  | 13.1  | 10.5  | (32) |  |
|                                                                        | 10%         | WB          | 8.3   | 8.4   | 10.2  | 12.3  | 15.5  | 17.6  | 18.8  | 18.6  | 17.3  | 14.6  | 11.0  | 8.6   | (33) |  |
|                                                                        |             | MCDB        | 9.1   | 9.5   | 12.2  | 15.7  | 19.5  | 21.8  | 23.7  | 22.6  | 20.3  | 16.1  | 12.0  | 9.3   | (34) |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 4.7   | 5.7   | 7.7   | 9.4   | 9.5   | 9.8   | 9.6   | 9.5   | 9.2   | 7.3   | 5.5   | 4.5   | (35) |  |
|                                                                        |             | MCDBR       | 5.1   | 6.9   | 11.1  | 13.4  | 13.7  | 13.6  | 13.6  | 14.1  | 12.7  | 8.3   | 6.2   | 5.0   | (36) |  |
|                                                                        | 5% WB       | MCWBR       | 4.7   | 5.3   | 7.0   | 7.3   | 7.3   | 6.8   | 5.9   | 6.1   | 6.4   | 5.3   | 5.0   | 4.8   | (37) |  |
|                                                                        |             | MCWBR       | 5.0   | 5.7   | 9.3   | 11.6  | 12.1  | 11.8  | 11.8  | 12.1  | 10.8  | 7.3   | 5.8   | 4.8   | (38) |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        | taub        | 0.327 | 0.348 | 0.384 | 0.410 | 0.420 | 0.418 | 0.418 | 0.408 | 0.382 | 0.364 | 0.338 | 0.322 | (40) |  |
|                                                                        |             | taud        | 2.393 | 2.357 | 2.247 | 2.191 | 2.207 | 2.238 | 2.238 | 2.289 | 2.348 | 2.397 | 2.398 | 2.367 | (41) |  |
|                                                                        | Ebn at Noon | Ebn at Noon | 704   | 781   | 811   | 830   | 838   | 842   | 834   | 824   | 806   | 751   | 681   | 646   | (42) |  |
|                                                                        |             | Edh at Noon | 65    | 85    | 112   | 132   | 136   | 133   | 131   | 119   | 101   | 81    | 64    | 58    | (43) |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      | RadAvg      | 0.84  | 1.54  | 2.72  | 4.29  | 5.00  | 5.50  | 5.22  | 4.45  | 3.41  | 2.00  | 0.99  | 0.67  | (44) |  |
|                                                                        | RadStd      | RadStd      | 0.09  | 0.28  | 0.36  | 0.48  | 0.47  | 0.51  | 0.51  | 0.43  | 0.27  | 0.17  | 0.10  | 0.09  | (45) |  |

## Historical Trends

|                              |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------------|--|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                              |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                              |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) Station Only            |  | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (63 neighbors) |  | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page